INTERNATIONAL MONETARY FUND

Are European Firms Financially Constrained?

Prepared by Luis Brandão-Marques, Alexandra Fotiou, Lea Havemeister, and Richard Varghese

WP/26/235

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2026
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WORKING PAPER

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IMF Working Paper
European Department

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Authorized for distribution by Malhar Nabar
October 2026

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RECOMMENDED CITATION: Luis Brandão-Marques, Alexandra Fotiou, Lea Havemeister, and Richard Varghese, "Are European Firms Financially Constrained?," IMF Working Paper WP/26/235.

JEL Classification Numbers:	G32, H25, D22
Keywords:	Financial constraints; leverage; corporate income tax; euro area firms; debt test
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* The authors would like to thank Malhar Nabar, Gee Hee Hong, and participants at the European Commission, ECB, and IMF seminars, for their suggestions and comments. Can Ugur, Kayla Qin, and Morgan Maneely provided excellent research assistance.

Are European Firms Financially Constrained?*

Luis Brandao-Marques† Alexandra Fotiou‡ Lea Havemeister§ Richard Varghese¶

October 20, 2026

Abstract

We examine if European firms face binding financial constraints that explain Europe's persistent scale-up gap. Using corporate tax changes as exogenous shocks and a "debt test", we analyze 7 million predominantly private firms across 20 euro area countries (2000–2022). A higher corporate tax rate increases the value of interest deductibility and, absent frictions, induces firms to increase leverage. We find that firms generally increase leverage after tax hikes, indicating constraints are not broadly binding. However, young firms and those in innovation-intensive industries show weaker responses. Among innovative firms, constraints vary: young firms appear unconstrained, while small, mature ones face binding constraints. Our findings suggest that financing frictions in Europe are concentrated among specific firm types at particular development stages, indicating that they are important at the critical scale-up stage.

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1 Introduction

Europe's productivity slowdown and per capita income gap relative to the U.S. have placed private investment and firm growth at the center of the policy agenda. In many European economies, business dynamism remains weak, young firms with high growth potential account for a smaller share of activity than in other advanced economies, and investment in innovation and intangible capital is low (Adilbish et al. 2025). Recent high-level reports underscore the need for structural reforms to unlock the region's economic potential that range from domestic reforms to improve the functioning of labor and product markets, deepening the European single market, and creating a savings and investments union (SIU) that can mobilize Europe's vast savings toward productive investment (Draghi 2024, Letta 2024, Arnold et al. 2025, Brandao-Marques et al. 2026). European policymakers point to financing frictions—amplified by bank dependence, shallow equity markets, and incomplete capital market integration–as a key bottleneck, reflecting concerns at the core of the capital markets union agenda (e.g., Pekanov 2024, Arampatzi et al. 2025). These concerns have shaped new initiatives ranging from the SIU to targeted public support for risk capital. Yet, a fundamental empirical question at the core of this debate remains insufficiently addressed: are European firms financially constrained, and if so, which ones?

We answer these questions by integrating the literature on the measurement of firm-level financial constraints (Kaplan and Zingales 1997, Whited and Wu 2006) with that on the role of corporate taxation in shaping debt capacity and capital structure (Desai et al. 2004). In the trade-off theory framework, a higher corporate tax rate increases the value of interest deductibility and, absent frictions, induces firms to increase leverage (Miller 1977).[1] We implement a debt-capacity-based test of financial constraints following Heider and Ljungqvist (2015) and use corporate tax changes as plausibly exogenous shifts in incentives to use debt finance. This approach exploits cross-country tax variation and allows us to assess firms' ability to expand leverage in response to changes in the marginal incentives to borrow. The debt test is most informative about frictions affecting firms' access to debt finance. Financing frictions can be multidimensional and need not coincide across debt, equity, and internal funding sources (Hoberg and Maksimovic 2015). For example, a muted leverage response may reflect the wish to preserve debt capacity to fund future investment projects

[1] Static trade-off theory implies immediate leverage adjustment; dynamic trade-off theory incorporates adjustment costs and frictions, so post-tax-hike leverage changes also reflect debt capacity and constraints (Leary and Roberts 2005).

(i.e., financial flexibility) when equity issuance is costly (Li et al. 2016), not necessarily a firm not being able to increase borrowing. We run this assessment for a large, harmonized panel of over 7 million predominantly private euro area firms and exploit national corporate income tax changes across 20 euro area countries over the period 2000–2022 as (plausibly) exogenous shocks to the demand for debt. To our knowledge, this is the first study to measure financial constraints at scale in Europe using a plausible identification scheme, thus enabling a direct assessment of debt capacity for millions of firms rather than the smaller subset of public firms that have been commonly used in the literature.

We have five findings. First, on average, firms systematically increase leverage following tax increases, consistent with the debt-test intuition and indicating that, on average, debt-related financing constraints are not broadly binding for European firms. This response is economically meaningful and driven by private firms that dominate the euro area economy: private firms increase long-term leverage by about 8 basis points for each one-percentage-point rise in the corporate tax rate, implying roughly a 0.9 percent increase in long-term debt levels following an average tax increase of 2.1 percentage points. Second, listed firms do not seem to be less financially constrained than private firms, which points to informational asymmetries not being a key driver of financial constraints among euro area firms. Third, young firms seem to be more constrained than older firms in accessing debt finance. Fourth, small firms are not as constrained as the rest, which suggests that firm size is a poor proxy for financial constraints. Finally, firms operating in more innovation-intensive industries exhibit weaker leverage responses to tax increases, consistent with tighter debt-financing frictions in these sectors. Moreover, financing constraints in innovative activities are nonlinear and point to a scale-up bottleneck: leverage responses are attenuated in innovation-intensive country–industry pairs, particularly among small firms, while young innovative firms (including young–small firms) remain able to expand borrowing when incentives strengthen. This pattern suggests that financing frictions in innovative environments are concentrated among small but more mature firms, which are past the startup phase and for whom financial constraints become binding precisely at the stage of their life-cycle when external finance is needed to scale.

Our findings are consistent with recent European evidence based on alternative data sources and approaches (e.g., survey-based and balance-sheet indicators), which points to financial constraints being significantly tighter for younger firms and those operating in innovation-intensive activities

than for the average firm (Asdrubali et al. 2022). Consistent with this, Amundsen et al. (2026) show that declining business dynamism has reduced the share of financially constrained firms over time by reducing the share of young firms in the data. Even in periods of relatively accommodative aggregate credit conditions, financing gaps persist for innovative firms and firms undergoing fast growth, particularly when funding needs are biased toward equity instead of standard bank debt (European Investment Bank 2023, Fratto et al. 2024). Relative to this literature, our contribution lies in a better causal identification since we implement a tax-based debt-capacity "debt test", covering millions of predominantly small, private firms. Crucially, we uncover a novel nonlinear pattern in innovation-driven sectors in which financing constraints are most acute among small but more mature firms, pointing to a missed scale-up opportunity that warrants policy attention.

This paper relates to literature on the measurement of firm-level financial constraints, which struggles with credibly identifying financially constrained firms. This is because firm-level financial constraints are not directly observable and empirical work must rely on proxies for the shadow cost of external funds. Canonical measures include investment–cash flow sensitivities (Fazzari et al. 1988), reduced-form indexes that combine observable firm characteristics (Kaplan and Zingales 1997, Whited and Wu 2006, Hadlock and Pierce 2010), and survey-based indicators of financing obstacles or credit rationing (Beck et al. 2005). Although influential, these approaches face well-known identification difficulties (see Farre-Mensa and Ljungqvist 2016).[2] The "debt test" approach that we follow has three advantages: it measures access to finance directly instead of endogenous outcomes like investment; it supports a causal interpretation because the shift in the demand for debt (i.e., the change in the tax rate) is plausibly exogenous; and it can be applied to public and private firms using data from commercially-available data sources like Orbis.

Our paper also relates to the literature on the role of corporate taxation in shaping debt capacity and capital structure, and extends the "debt test" logic of Heider and Ljungqvist (2015) to cover the predominant form of firm in Europe—small and unlisted companies. In the classic trade-off framework, interest deductibility generates a debt tax shield; higher statutory corporate tax rates raise the value of deductions and should increase optimal leverage absent binding frictions

[2]Investment-cash-flow sensitivities can load on unobserved investment opportunities, adjustment costs, and measurement error in Tobin's q, so high sensitivity need not imply tight finance (Kaplan and Zingales 1997). Index-based measures map from mostly U.S. public firm characteristics to constraints and may primarily capture life-cycle stage or risk rather than marginal financing frictions (Farre-Mensa and Ljungqvist 2016, Hadlock and Pierce 2010). Survey-based measures may be affected by perception biases and endogenous demand for external finance (Beck et al. 2005).

(Modigliani and Miller 1963, Graham 2000). Empirically, staggered tax changes have been shown to move leverage in economically meaningful ways and to do so asymmetrically, with stronger responses to tax increases than to tax cuts (Heider and Ljungqvist 2015). However, the evidence is mixed once we consider small and private firms. For example, Ivanov et al. (2025) find that small, private firms in the U.S. increase leverage after a tax cut because the latter increases the return on capital, but Faccio and Xu (2025) find the opposite. We add to this literature because Europe is a very informative setting due to the cross-country heterogeneity in corporate tax systems, the prevalence of closely-held firms, and the limited access to public equity markets. These factors amplify the role of debt and the consequences of debt-biased taxation (Huizinga et al. 2008).

The remainder of the paper proceeds as follows. Section 2 describes the data sources and presents summary statistics. Section 3 outlines the empirical strategy and identification. Section 4 presents the benchmark leverage responses, explores heterogeneity by firm type and sectoral innovation intensity, and reports robustness checks. Section 5 concludes with policy implications for corporate taxation and the financing of European scale-ups.

2 Data

Our analysis integrates data from multiple sources. We combine firm-level balance sheet data for euro area firms with national corporate income tax changes for 20 euro area countries over the period 2000–2022. We supplement this dataset with country-level macroeconomic variables to control for business cycle conditions and other country characteristics. Finally, we incorporate a series of indicators to examine sectoral and country-specific factors that may amplify or mitigate our findings. This section provides an overview of the data sources and variables and discusses the sample characteristics used in our empirical analysis.

2.1 Firm-level Data and Variables

We obtain firm-level information from Bureau van Dijk's Orbis Historical, a comprehensive data source for European firms. We combine Orbis's datasets on firmographics and financials for listed and unlisted firms across 20 euro area countries over the period 2000–2022. Appendix A details the data cleaning steps. The final dataset includes firm-level characteristics such as age, location,

listing status, and sector, alongside detailed balance sheet information about assets, liabilities, and key financial ratios.

Table 1 reports summary statistics for unlisted firms, which represent the majority of observations in our sample. The median unlisted firm in our sample is relatively small, with total assets of approximately $0.5 million and 4 employees, a median age of 11 years, and profitable, exhibiting a return on assets (ROA) of 2.5 percent. These firms also maintain conservative capital structures, with a median overall leverage ratio of 8.3 percent. The mean values for all these variables are noticeably higher than the medians, indicating a right-skewed distribution. Table B.2 presents summary statistics for the full sample. These closely mirror those for unlisted firms, reflecting the dominance of unlisted firms in the sample.

Table 1: Summary Statistics: Unlisted Euro Area Firms

Unlisted firms only.	**Min**	**p25**	**Mean**	**Median**	**p75**	**Max**	**SD**	**Obs**
Age	2.00	6.00	13.66	11.00	19.00	61.00	10.36	47.86
Employees	0.00	2.00	14.28	4.00	10.00	736489.00	237.98	30.35
Total assets (mio.)	0.00	0.14	49.07	0.49	1.65	10954.44	498.47	47.86
Asset growth (%)	-1103.77	-15.10	-0.64	-1.29	14.60	957.05	39.09	38.13
EBITDA (mio.)	-21357186.43	0.00	3.81	0.03	0.11	12918718.98	4595.46	38.50
Δ Cash/Assets (pp)	-417784.60	-3.61	-0.03	0.02	4.42	32391.26	127.95	38.98
Tang. FA (TFA, mio.)	0.00	0.01	10.93	0.05	0.32	476013.03	202.84	46.06
TFA/Assets (%)	0.00	2.57	24.73	12.73	39.08	4288791.41	632.52	46.06
Intangible FA (mio.)	0.00	0.00	2.19	0.00	0.01	10300	45.1	46.06
Intangible FA/Assets (%)	0.00	0.00	4.11	0.00	0.74	6551.93	12.82	46.06
Investment (%)	-5954583.50	-5.46	-4.53	-0.57	2.12	36343.08	1061.24	47.86
Sales (imp., mio.)	0.00	0.13	179.37	0.58	4.89	4097116.47	1773.52	47.86
Sales growth (imp., %)	-100.00	-16.67	16.84	-0.51	17.36	3231.20	121.83	47.86
Turnover (mio.)	-9383.36	0.11	77.68	0.38	1.43	4097116.48	1807.52	40.66
ROA (%)	-100.00	-0.90	3.80	2.53	9.95	100.00	19.44	40.49
Long-term debt (mio.)	0.00	0.00	10.98	0.01	0.18	21847.84	213.17	39.48
Long-term lev. (imp., %)	0.00	0.00	18.88	9.79	29.13	212.08	25.09	47.86
Δ Long-term lev. (%)	-212.07	-1.77	-0.22	0.00	0.00	212.08	14.54	38.11
Leverage (%)	0.00	0.00	21.48	8.33	32.97	6225326.17	1466.07	36.74
WW index	-133.18	-2.16	-1.52	-0.77	0.43	33.85	5.66	47.86

This table shows the summary statistics for key characteristics (age, number of employees), balance sheet variables with a particular focus on the debt-related variables, and one traditional index of financial constraints from the literature for unlisted firms. FA refers to fixed assets; TFA to tangible fixed assets; ROA to the return on assets. WW index refers to the Whited-Wu (2006) index, which takes higher values for more constrained firms. Observations are reported in millions.

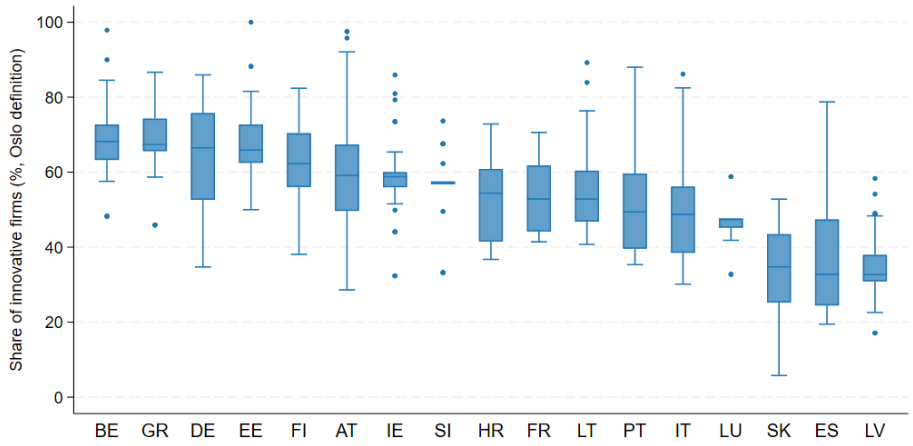
Next, we augment our database with a set of firm and country characteristics that could mitigate or amplify our findings on financial constraints facing euro area firms. First, we identify venture-capital-backed firms in our sample using a dummy variable based on the VICO dataset, which contains detailed geographical, industry, and accounting information on start-ups that received at least one venture capital investment between 1998 and 2021 (see Bertoni and Martí (2011), Bertoni et al. (2015)). Second, we use the OECD's Oslo innovation indicator which varies across country, sector, and time to examine if there is a difference in financial constraints faced by more innovative firms (see OECD and Eurostat 2018).[3] The OECD reports the share of innovative firms

[3]See Business innovation statistics and indicators for details.

according to four harmonized indexes for country-sector pairs at the two-digit NACE2 sector level, which are updated every two years. We use the narrow measure of innovation according to the Oslo Manual. A country-sector pair is classified as innovative if it exhibits an innovative firm share in the highest quartile in a given year.

Figure 1 documents substantial heterogeneity in innovation intensity across both countries and sectors, measured by the OECD share of innovative firms (Oslo definition).[4] This two-dimensional variation—across country-level medians and within-country sectoral dispersion—ensures that innovation intensity is not driven by a single sector or national profile. Consequently, the data provide ample variation to implement panel specifications with rich fixed effects while retaining identifying power from policy-induced shifts in financing incentives interacting with heterogeneous firm environments.

Figure 1: Distribution of Innovation Shares Across Countries



The figure plots he interquartile range of the distribution across sectors of the innovation variable (i.e., the share of innovative firms in a sector, for each country) based on the OECD's Oslo innovation indicator. Values are ordered by the median innovation share of all sectors in a country. We exclude from the chart Cyprus and Malta because there is no data on the OECD indicator, and the Netherlands because it reports the same value for all sectors.

2.2 Composite Corporate Tax Rate Indicator

We merge the firm-level data with changes in tax rates obtained from the OECD Corporate Income Tax Rates Database. We construct a consistent series of corporate tax measures for the period 2000–2022, drawing on multiple indicators. For the years 2000–2017, the series is based on

[4]Examples of highly innovative country-sector pairs include manufacturing, chemicals, and knowledge-intensive services such as telecommunications in countries including Germany, Austria, Belgium, Finland, and Greece. Because the Oslo framework captures both product and process innovation, some less obvious sectors, including utilities and water management activities, can also be classified as highly innovative in specific countries and years.

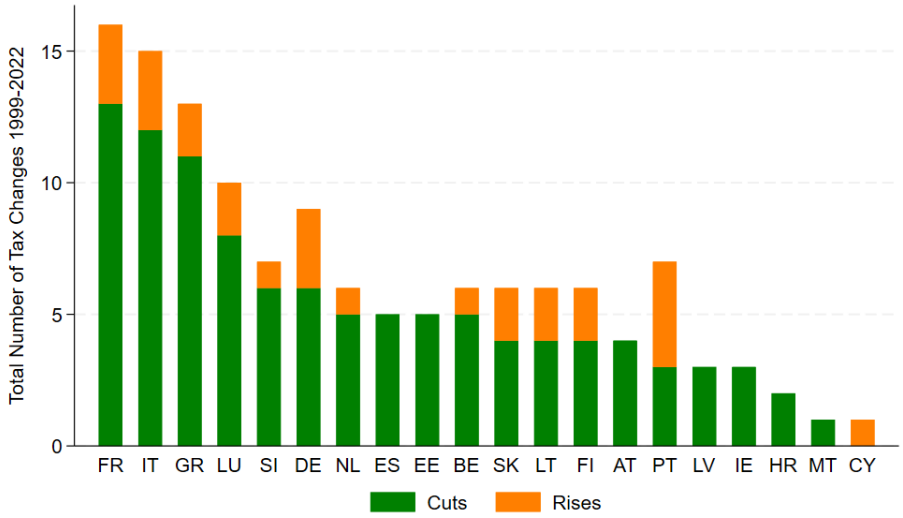
statutory corporate income tax rates (CIT) augmented with capital allowances, which captures the headline tax rate and key deductions available to firms. From 2018 onward, we utilize the composite effective average tax rate (EATR) provided by the OECD, which incorporates ten distinct tax characteristics—including CIT, allowances, and other relevant provisions—to provide a more comprehensive measure of the tax burden faced by corporations. Using effective tax rates is preferable to relying solely on statutory rates, as the EATR accounts for the impact of deductions, exemptions, and other tax features that materially affect firms' actual tax liabilities. This approach provides a more accurate representation of both intertemporal variation and cross-country comparability in corporate taxation—the latter being particularly critical for the euro area.[5]

Figure 2 shows substantial cross-country heterogeneity in the measure of corporate income tax changes. Tax cuts are more frequent than tax increases: in total, 104 tax cuts affect around 4.8 million firms in our sample, whereas only 27 tax rises affect approximately 2.7 million firms, with seven countries experiencing no tax increases over the sample period. The magnitude of tax changes is also asymmetric. The average tax increase amounts to 2.1 percentage points, ranging from marginal adjustments of 0.01 percentage points to hikes as large as 10 percentage points. By contrast, tax cuts are larger on average—around 3.0 percentage points—and exhibit substantially greater dispersion, with reductions of up to 15.8 percentage points. This asymmetry underscores a persistent policy bias toward tax reductions in recent decades (Garcia-Bernardo et al. 2022), as further illustrated in Figure 3, which depicts the evolution of tax changes over time. This pronounced variation across countries, and between tax increases and cuts, provides rich quasi-experimental variation that we exploit to identify firms' leverage responses to changes in the value of debt tax shields.

One consideration is that corporate income tax rates have predominantly declined over recent decades, implying fewer tax increases and therefore less variation on the intensive margin relevant for identifying leverage responses. Although this asymmetry may affect statistical precision, it does not materially affect the identification strategy because, as discussed later, it relies on the tax changes as exogenous shocks to the incentive to borrow, and the presence of multiple, sizable tax increases across countries and over time provides sufficient variation to implement the debt test.

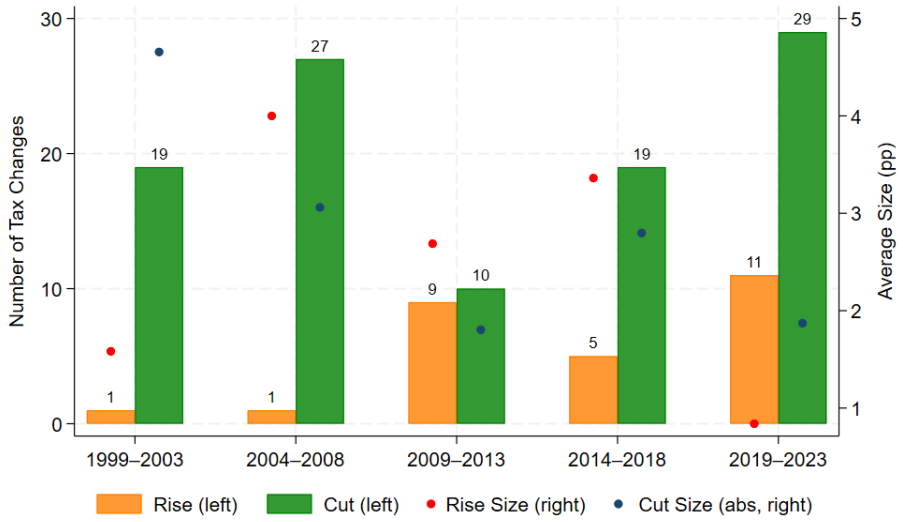
[5] Additional details on the distribution of observations across countries and years, as well as the country-year tax reform episodes underlying the identification strategy, are reported in Appendix Tables B.6-B.9. See also OECD Corporate Income Tax Rates Database for details.

Figure 2: Number of Tax Changes by Country



The figure plots the number of tax cuts (green) and tax rises (orange) that occurred between 2000-2022 by country.

Figure 3: Number of Tax Changes over Time



The figure plots the number of tax cuts (green bars) and tax rises (orange bars) by time period. The dots (right axis) show the average size of cuts (in absolute values, blue dots) and rises (red dots).

2.3 Macroeconomic Data and Variables

Finally, we incorporate country-level macroeconomic controls to account for broader economic conditions that may influence firm behavior. These controls include GDP growth, consumer price inflation (CPI), and the unemployment rate, which capture aggregate demand and labor market dynamics. The monetary policy stance is proxied by the policy rate, while financial conditions are reflected in 2-year sovereign yields and 10-year sovereign yield spreads, indicating short-term and

long-term borrowing costs and sovereign risk premia. Fiscal policy enters our specification through government expenditure and government revenue excluding corporate income tax (CIT), providing a measure of public sector demand and fiscal space. All macroeconomic variables are sourced from the IMF World Economic Outlook (WEO), while the sovereign yields are sourced from Bloomberg and Haver Analytics.[6] Table B.3 reports the summary statistics for the tax changes and other macro control variables.

3 Empirical Strategy

Absent taxes and other frictions, Modigliani and Miller's (1958) theorem implies that a firm's capital structure is irrelevant for its value. Introducing corporate taxes, however, provides an incentive to increase leverage to benefit from the deductibility of interest payments (Modigliani and Miller 1963, Miller 1977). In a static trade-off theory framework, firms will target an optimal level of debt that balances the increased value of interest deductibility brought by a higher corporate tax rate against bankruptcy costs associated with higher leverage. While static trade-off theory predicts an immediate adjustment to a new optimal capital structure (holding expected distress costs fixed), dynamic trade-off theory emphasizes that adjustment costs and financial frictions cause rebalancing to be gradual (Leary and Roberts 2005). Consequently, the measured increase in leverage after a tax hike reflects not only the strength of tax incentives but also the interaction between those incentives and firms' debt capacity. Financially unconstrained firms respond more strongly and quickly than those facing tight borrowing limits because of weak collateral, short credit histories, or informational frictions, which lead to muted or delayed adjustments. Trade-off theory, thus, links taxation to leverage changes, delivering a transparent empirical test of financing constraints that are relevant for Europe's predominantly unlisted and bank-reliant corporate sector.

Our empirical approach builds on this insight that there is a tax advantage (i.e., tax shield) for firms to rely on debt rather than equity financing. We follow Heider and Ljungqvist (2015) who exploit exogenous U.S. state-level corporate income tax increases as shocks to the value of tax shields and examines firm-level leverage adjustments in response. Firms that significantly increase their leverage following a tax hike are classified as unconstrained, reflecting their ability to access

[6] See IMF World Economic Outlook Database for details.

debt financing when incentives to borrow strentghen. In contrast, firms that do not adjust their leverage are either financially constrained or unresponsive for idiosyncratic reasons. Similarly, we use national corporate income tax changes in the euro area as exogenous shocks to the demand for debt to evaluate whether euro area firms face financial constraints.

We estimate the response of firm-level leverage to lagged changes in our composite corporate tax rate measure while controlling for firm fundamentals and macroeconomic and financial conditions. We also include firm fixed effects (to control for unobservable firm characteristics) and industry-year fixed effects (to control for demand shocks and preferential tax treatments at the sector level) in our benchmark specification, and allow different leverage responses to tax increases and tax cuts. Our framework accommodates repeated tax changes and can be interpreted as a difference-in-differences regression under the assumption that, absent tax changes, the leverage of treated firms (those that experience tax hikes) would have followed the same trend as that of control firms (i.e., those that do not). Identification comes from within-firm variation in leverage around policy-driven changes in the value of the debt tax shield. Farre-Mensa and Ljungqvist (2016) call this approach to assessing financial constraints a "debt test".[7]

To estimate firms' leverage responses to tax changes, we employ a panel regression framework at the firm-year level. Specifically, for firm i in industry j and country c in year t, we model the change in long-term leverage ratio as follows:

$$\Delta Lev_{ijct} = \beta_1 \cdot rise_{ct-1} + \beta_2 \cdot cut_{ct-1} + \gamma \Delta X_{it-1} + \delta \Delta Z_{ct-1} + \alpha_i + \theta_{jt} + \epsilon_{ijct} \tag{1}$$

where ΔLev_{ijct}, our dependent variable, denotes the change in the long-term leverage ratio, calculated as the difference in the ratio of long-term debt to total assets between years t and $t-1$ as given below.[8]

[7] The "debt test" offers a transparent approach to identifying financially constrained firms by examining whether they increase leverage in response to tax-induced changes in the value of the debt tax shield, and hence in their incentives to borrow. This approach circumvents the limitations of traditional measures that may conflate financial constraints with firm life-cycle characteristics or investment opportunities (Heider and Ljungqvist 2015, Farre-Mensa and Ljungqvist 2016), and abstracts from other confounding factors unrelated to financing constraints, such as relationship lending.

[8] We use long-term leverage because it likely is more sensitive to tax changes than short-term debt, which is often used for working capital requirements (Farre-Mensa and Ljungqvist 2016). Long-term debt also generates larger debt shields as the loan size is usually larger.

$$\Delta Lev_{ijct} = \frac{\text{Long-termDebt}_{it}}{\text{TotalAssets}_{it}} - \frac{\text{Long-termDebt}_{it-1}}{\text{TotalAssets}_{it-1}} \tag{2}$$

The variable $rise_{ct-1}$, our main explanatory variable of interest, captures continuous positive changes in the national corporate tax measure with β_1 quantifying the leverage response to tax increases. Conversely, cut_{ct-1} reflects continuous negative changes in the tax measure, and β_2 measures the leverage response to tax cuts.

Firm-level controls (X_{it-1}) include changes in firm size (total assets growth) and return on assets, which control for changes in investment opportunities. Country-level controls (Z_{ct-1}) include GDP growth, consumer price inflation (CPI), the unemployment rate, two-year government bond yields, ten-year yield spreads, and the logarithms of government expenditure and revenue (excluding CIT receipts), which mitigate confounding effects from macroeconomic developments and omitted variable bias. The specification incorporates firm fixed effects (α_i) and industry-year fixed effects (θ_{jt}), with standard errors clustered at the country level.

Following Heider and Ljungqvist (2015), we estimate the model in first differences. This accommodates repeated treatments (i.e., allowing firms to experience multiple tax increases or cuts over the sample period), treatment reversals (e.g., a tax increase followed by a tax cut), and asymmetries in firms' responses to tax changes (i.e., controlling for tax cuts).

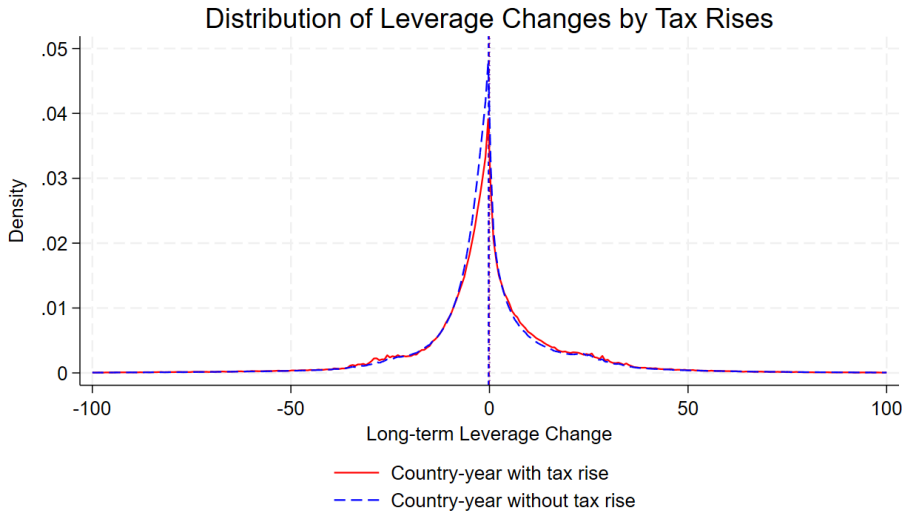
A key identifying assumption underlying our empirical strategy is that, absent a change in tax rates, the leverage of treated firms would have evolved in parallel with that of suitable control firms (i.e., parallel trends assumption). In other words, absent a tax change, firms' leverage would have continued to follow its prior trend. The empirical test is thus structured around β_1, the main coefficient of interest. The null hypothesis ($\beta_1 = 0$) posits that firms do not adjust their leverage in response to a tax increase, indicating that they are financially constrained or decide not to raise leverage for idiosyncratic reasons. A muted response may also reflect limited taxable profits, which reduce the value of additional interest deductions, or high expected distress costs that make firms reluctant to increase borrowing despite stronger tax incentives.The alternative hypothesis ($\beta_1 > 0$) suggests that firms significantly increase leverage following a tax rise, consistent with being financially unconstrained and able to exploit the enhanced value of debt tax shields.

4 Results

4.1 Distribution of Firm-level Changes in Leverage

We start by comparing the distribution of firm-level changes in leverage around corporate income tax increases using a simple nonparametric approach. Figure 4 shows that more firms increase leverage in a year with a tax increase compared with a year without one. The difference between the two distributions is statistically significant, as shown by the Kolmogorov–Smirnov (KS) test statistic of 0.0961, significant at all conventional levels. This evidence indicates that firms exposed to tax increases systematically raise their leverage, whereas firms in periods without tax changes exhibit little adjustment.

Figure 4: Distribution of Leverage Changes by Tax Rises



The figure plots the distribution of changes in the long-term leverage ratio between country-year observations with a tax rise and those without.

This pattern is consistent with the intuition underlying the "debt test". Financially unconstrained firms respond to higher tax rates by increasing debt to exploit the enhanced value of interest tax shields, whereas constrained firms are unable to do so. The resulting shift in the leverage distribution following tax hikes therefore provides a transparent and agnostic indicator of financial constraints—only firms with sufficient access to external finance adjust their capital structure in response to tax-induced changes in borrowing incentives.

4.2 Leverage Response to Tax Changes

We document European firms' leverage responses to tax changes by estimating our benchmark specification in Equation 1. The results are in Table 2. Column (1) shows the estimates for private firms of the specification that includes firm-level controls, macroeconomic controls, industry-year fixed effects, and firm fixed effects.[9] Column (2) does the same for public firms, and (3) presents estimates using the entire sample of private and public firms of a specification that augments the benchmark regression with an interaction between tax changes and a dummy variable for public firms.

The estimates of β_1, our key coefficient of interest, are positive and statistically significant. Firms significantly raise leverage in response to tax hikes, consistent with them taking advantage of the increased value of debt tax shields. The results for the entire sample and for the unlisted firms only are the same and listed firms do not have a different response. In our sample, on average, firms increase leverage by 8 basis points after a one-percentage point increase in the corporate income tax rate (Column 1, Table 2). The results for listed firms only, in Column (2), shows that the response of listed firms, although larger in magnitude than that of private firms, is statistically significant only at lower confidence levels. These estimates are in the lower range of estimates found in the literature (see Table B.1), which range between zero and 140 basis points.[10]

These results have three main implications. First, they suggest that financial constraints are not binding for the average European firm and the average firm in the euro area retains some access to debt finance when borrowing incentives change. It is possible that the average firm in our sample would have preferred to have increased leverage more in response to the tax hike, and was not able to because at some point the borrowing constraint became binding. In that sense, our result could be understating the importance of binding borrowing constraints at the firm level. However, following Li et al. (2016), firms' optimal leverage balances the benefit of maintaining financial flexibility (given costly equity issuance, borrow less know to be able to borrow more in the future) and benefit of the tax advantage of debt. They estimate that firms will refrain from increasing leverage in response to a tax increase even if the borrowing constraint is not binding

[9] The results are robust to different assumptions regarding unobserved heterogeneity, including different combinations of firm-level controls, industry-year fixed effects, and country fixed effects.

[10] Only one study that we surveyed finds a significant negative response for small private firms, which cut leverage by 4 percent in response to a tax hike (Ivanov et al. 2025).

Table 2: Leverage Response to Tax Changes

Sample	(1) Unlisted	(2) Listed	(3) All	(4) Unlisted	(5) All
Rise	0.080*	0.129	0.080*	0.075*	
	(0.041)	(0.151)	(0.041)	(0.037)	
Rise × Listed			0.091		
			(0.095)		
WW				-0.077***	
				(0.020)	
Rise × WW				0.036	
				(0.031)	
Rise (small)					0.627
					(0.373)
Rise (medium)					0.423**
					(0.182)
Rise (large)					0.293**
					(0.112)
Cut	-0.007	0.273***	-0.007	0.001	
	(0.032)	(0.087)	(0.032)	(0.031)	
Cut × Listed			0.119***		
			(0.033)		
Cut × WW				-0.026**	
				(0.012)	
Cut (small)					-0.146
					(0.312)
Cut (medium)					0.497
					(0.390)
Cut (large)					-0.087
					(0.099)
Observations	24,338,530	4,280	24,347,346	24,338,530	24,347,346
R-squared	0.115	0.221	0.115	0.115	0.115
Firm controls	✓	✓	✓	✓	✓
Macro controls	✓	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓

This table reports coefficients from benchmark panel regressions of the change of firms' long-term leverage ratios on corporate tax changes. Columns (1)–(4) report results for the unlisted, listed, full, and unlisted-firm samples, respectively. Note that the difference in sample sizes between the sum of columns (1) and (2), and column (3) is due to the inclusion of delisted firms in (3), and a different number of singleton observations. Column (3) includes interactions with listing status, and column (4) includes interactions with the Whited-Wu indicator. Column (5) replaces the continuous rise and cut measures with indicators for small, medium, and large tax changes based on terciles of the respective distributions. Specifically, for rises and cuts separately, we split the distribution of tax changes into terciles of the country-year distributions of rises and cuts, and construct three indicators per direction, giving six mutually exclusive dummies that enter the regression jointly. Country-years with no tax change are the omitted baseline, so each coefficient measures the average leverage response (in percentage points) to a tax change of the relevant size class relative to no change. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding corporate income taxes. All independent variables are lagged by one period. The models include industry-year and firm fixed effects, with robust standard errors clustered at the country level. Significance levels are * p<0.10, ** p<0.05, and *** p<0.01.

because of the benefit of financial flexibility (i.e., they will maintain a leverage buffer). Applying this reasoning to our setting makes our findings more striking, as it means that, if anything, we could be understating our case that financial constraints are not binding for the average euro area firm.

Second, the estimates of β_2, capturing the response to tax cuts, are negative but not statistically significant in our main specifications for unlisted firms and the full sample. Like Heider and Ljungqvist (2015), we interpret this as consistent with the dynamic trade off theory's prediction that tax increases and decreases need not have symmetric effects, and that the response should be more significant for tax increases.[11] Empirical studies in corporate finance consistently document that firms tend to increase leverage following tax hikes, as higher tax rates enhance the value of interest deductibility and incentivize debt financing (see Table B.1). Notably, Heider and Ljungqvist (2015) find that firms increase leverage by approximately 40 basis points in response to state-level tax increases. This tax sensitivity is greater among profitable and investment-grade firms, which respectively benefit from a higher marginal tax advantage and face a lower marginal cost of issuing debt. In contrast, the leverage response to tax cuts is less clear-cut: Some studies report a decrease in leverage (Devereux et al. 2018, Fleckenstein et al. 2020, Faccio and Xu 2025) consistent with an expectation of symmetric leverage response to tax changes, while others find no significant response (Goodman and Isen 2025), or an increase in leverage following tax reductions (Ivanov et al. 2025, Cui et al. 2025).

Third, European private firms are not systematically more financially constrained than their public counterparts. European public firms increase debt by 13 basis points in response to a one-percentage point increase in the corporate income tax rate (Column (2), Table 2), but this is only about half of what is reported by Heider and Ljungqvist (2015) for US listed firms.[12] The leverage response to corporate tax changes does not differ by firms' listing status because Column (3) of Table 2 shows that the interaction coefficient of the tax increase and listed status dummies is not statistically significant. Thus, the positive and statistically significant response of leverage to tax increases is driven primarily by private firms (the vast majority of firms in our sample). Given that

[11] This is because the benefits of reducing debt after a tax cut accrue mostly to debtholders, while reducing the value of shareholders' option to default (see Heider and Ljungqvist 2015).

[12] Although not the central object of this paper, the difference between euro area and US firms' debt response to tax hikes could be explained by the former being more opaque than the latter (e.g., because of regulatory fragmentation or different ownership structures) and, thus, more susceptible to asymmetric information, or having lower financial strength.

private firms are, on average, more opaque than public ones, this result suggests that differences in asymmetric information may not be the main driver of financial constraints among European firms.

These results are also economically significant. Quantitatively, unlisted firms increase long-term leverage by about 8 basis points for each one-percentage-point increase in the corporate tax rate (Table 2, Column (1)), corresponding to a 0.9 percent increase in long-term debt levels following an average tax rise of 2.1 percentage points. In monetary terms, this implies an increase in debt of approximately $82,300 for the average private firm (or 0.2 percent of assets),[13] assuming total assets remain unchanged.

Since much of the existing empirical evidence on financial constraints is based on samples of publicly listed firms, predominantly in the United States, we compare the performance of Heider and Ljungqvist's debt test measure to Whited and Wu's index (WW), a widely used measure of financial constraints. We do this by including an interaction of tax changes dummies with another dummy variable that is equal to one for firms classified as financially constrained—defined as having a WW index in the highest tercile (and zero otherwise).[14] This comparison is important because indexes like WW, although easy to use (unlike the debt test), were developed and validated largely in the context of listed US firms and their ability to estimate financial constraints may be weaker for other types of firms, including private euro area companies.

Column (4) in Table 2 shows the comparison between the debt test measure and the WW index. It shows that the interaction of tax changes with the Whited–Wu index is not significant. Hence, firms classified as constrained by the WW index have the same leverage responses to tax increases as the average firm (i.e., they do not seem to be financially constrained). This result highlights the challenges of directly applying classification schemes calibrated on U.S. public firms to samples composed largely of private firms (like the euro area) and favors using the debt-test approach in

[13] As mentioned before, the distribution of assets, sales, long-term debt, and leverage are very skewed. Total assets of the median firm are only $0.5 million, which compares to $49.1 million for the average firm.

[14] The higher the WW index value, the more constrained the firm, according to Whited and Wu (2006). In line with the existing literature, we take the coefficients from the original paper and construct the index as a linear combination of balance sheet variables. Specifically, for firm i in sector s at time t, we construct

$$WW_{it} = -0.091 * CF_{it} - 0.062 * listed_{it} + 0.021 * Lev_{it} - 0.044 * \log(TA)_{it} - 0.035 * SalesGr_{it} + 0.102 * IndSalesGr_{st}, \quad (3)$$

in which *CF* is cash flow, *Lev* is the long-term leverage ratio, *TA* is total assets, *SalesGr* is annual firm sales growth, and *IndSalesGr* is the industry's average annual sales growth.

this context, as it relies directly on firms' observed financing behavior in response to exogenous changes in borrowing incentives.

We next examine whether the average leverage response of unlisted firms could depend on the size of the change in the corporate tax rate. This is important because of the asymmetric distribution of corporate tax changes in our sample, in which large tax hikes are relatively rare and small tax changes may be less salient for firms' financing decisions. Moreover, testing if leverage reacts more to large than small tax hikes informs about corporate tax changes having threshold effects, possibly because of transaction or adjustment costs. It can also tell us whether standard linear estimates are understating the real response. Column (5) in Table 2 reports the results of this analysis for the full sample of firms.[15] Our results indicate that the positive average response of leverage to tax increases is primarily driven by medium-sized and large tax rises, rather than very small tax changes. This finding suggests that firms respond most strongly to tax changes that are sufficiently large to meaningfully alter borrowing incentives. This finding is consistent with Heider and Ljungqvist's (2015).

The finding that the average European firm is not financially constrained is particularly informative in the context of the European policy debate, which often attributes firms' limited ability to scale up to pervasive financing constraints, among other factors (Draghi 2024, Letta 2024, Fratto et al. 2024, European Commission 2025, Baumann et al. 2026, Evroux et al. 2025). Although concerns about bank dependence and underdeveloped capital markets feature prominently in policy discussions, our results point to a more nuanced interpretation. At the same time, aggregate responses may conceal substantial heterogeneity. Financing constraints may remain binding in particular environments or for specific groups of firms, motivating a closer examination of differential leverage responses. Therefore, in the next section we assess whether the financial constraints measured by firms' borrowing response to tax increases depend on observable firm characteristics.

4.3 Leverage Response by Firm Type

Firm size and age are commonly used in the literature as proxy indicators of the likelihood that a firm is financially constrained, as they capture systematic differences in information frictions, collateral availability, and credit history, all of which influence access to external finance. Leverage

[15] For unlisted firms, the estimates are virtually identical. Available on request.

is also often used to proxy financial constraints, but leverage is an equilibrium outcome (as implicit in the approach in use in this paper), not a direct measure of the tightness of a firm's financing constraint (Catherine et al. 2022). We examine whether leverage responses to tax changes vary systematically along firm size and age and modify the benchmark specification 1 by introducing an interaction of the tax variable, *rise*, with two dummy variables representing young and small firms, *Young* and *Small*, respectively. The specification becomes:

$$\Delta Lev_{ijct} = \beta_1 \cdot rise_{ct-1} + \beta_2 \cdot Type_{ijc,t-1} + \beta_3 \cdot rise_{ct-1} \times Type_{ijc,t-1} + \gamma \Delta X_{it-1} + \delta \Delta Z_{ct-1} + \alpha_i + \theta_{jt} + \epsilon_{ijct}, \quad (4)$$

where $Type \in \{Young, Small\}$. For this analysis, firms are classified as young if they are younger than seven years ($Young_{ijc,t-1} = 1$), which is the bottom quartile of the age distribution, and as small if their total assets fall within the bottom quartile of the size distribution ($Small_{ijc,t-1} = 1$). Table B.4 compares summary statistics for young firms, with a median age of four years, to those for old firms, with a median age of fifteen years. Young firms are on average smaller than older firms with about one-third of the total assets of old firms. Young firms have significantly more intangible assets (as a share of total assets) than old firms (6.4 vs. 3.2 percent). This suggests that young firms are less likely to have sufficient collateral to access credit since intangible assets, compared to physical assets, make poor collateral (Titman and Wessels 1988, Hale and Santos 2009). At the same time, they display higher asset and sales growth, investment rates, and returns on assets, consistent with faster growth dynamics.

As shown in table B.4, young and old firms have similar mean and median leverage ratios, but old firms show much more dispersion in their leverage ratios. The lower dispersion in leverage among young firms can be explained by them being more constrained than older firms. In models with limited commitment or enforcement, frictions like opacity, short credit histories, and lack of collateral dictate how much a firm can borrow and, because young firms have high growth and investment opportunities, they will borrow as much as they can (i.e., they will "hug the constraint"; see Li et al. 2016). Hence, while young firms with low net worth and good opportunities to grow will see their leverage ratios cluster close to where borrowing constraints are binding, older firms may decide to keep borrowing space to preserve financial flexibility or for other strategic reasons.

Table B.5 compares small and medium-and-large (i.e., larger) firms. Although the size classification is based on total assets, it maps closely into other measures of firm scale: small firms employ, on average, three workers, compared to eighteen workers for larger firms. Small firms tend to be younger with a median age of seven years, compared to twelve years for larger firms. They also exhibit lower asset growth and investment rates than larger firms. Small firms hold less fixed assets and less intangible fixed assets (as a share of total assets) than larger firms. And, although the average leverage of small and larger firms is similar, small firms have a lower median leverage ratio than larger firms, with more than half of them having no leverage. Similarly to young firms, and likely for the same reasons, the leverage ratio of small firms has much smaller dispersion than that of larger firms.

Table 3: Heterogeneity by Firm Type

Characteristic	(1) Young	(2) Small	(3) Young × Small	(4) VC
Young	-0.369*		-0.331	
	(0.195)		(0.194)	
Small		0.356**	0.445**	
		(0.134)	(0.161)	
Young × Small			-0.188	
			(0.125)	
Rise	0.088*	0.065	0.070	0.080*
	(0.045)	(0.041)	(0.043)	(0.041)
Rise × Young	-0.073***		-0.062***	
	(0.018)		(0.018)	
Rise × Small		0.058**	0.076*	
		(0.026)	(0.042)	
Rise × Young × Small			-0.050	
			(0.042)	
Rise × VC				-0.166***
				(0.021)
Observations	24,338,530	24,338,530	24,338,530	24,338,530
R-squared	0.115	0.115	0.115	0.115
Firm controls	✓	✓	✓	✓
Macro controls	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓

This table reports coefficients from the baseline panel regression for unlisted firms and various firm characteristics. The dependent variable is the change in the firm's long-term leverage ratio in percent. Rise is a continuous indicator of increases in our composite corporate tax rate measure and the main independent variables. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment rate, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding from CIT. All independent variables are lagged by one period. *Young* equals 1 if the firm is younger than 7 years, corresponding to the bottom age quartile. *Small* equals 1 if the firm's total assets are in the bottom quartile. *VC* equals 1 if the firm has received VC funding and is not time-varying. The empirical model is estimated in first differences using industry-year fixed effects, firm fixed effects, and robust standard errors clustered at the country level. Significance levels are * p<0.10, ** p<0.05, *** p<0.01.

Table 3 reports results from specifications that interact tax changes with indicators for firm age, firm size, and their interaction, as reported in Columns (1)-(3). Young firms exhibit a significantly weaker leverage response to tax increases relative to the average firm in our sample, as indicated

by the (statistically significant) negative coefficient of interaction between tax rises and the young-firm dummy (Column (1), Table 3): young firms increase their debt levels by about 1.5 basis points following a one-percentage-points increase in the corporate income tax rate, compared with an increase of 8 basis points for older firms. This finding is consistent with the more significant frictions faced by young firms, and make the borrowing constraints bind at lower levels of leverage, thus limiting their ability to adjust borrowing even when debt incentives strengthen. Figure 5 shows the estimate leverage response of young and old firms following 2.1 percentage-points increase in the corporate income tax rate (the sample average).

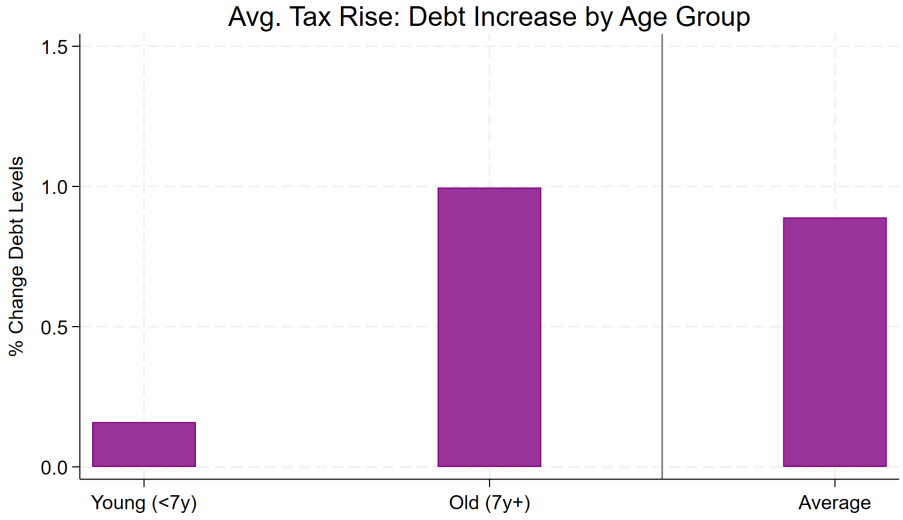
By contrast, small firms do not appear to be systematically constrained. The estimates in (Column (2), Table 3) show that small firms increase leverage by 12.3 basis points compared to 6.5 basis points for medium and large firms, in response to the same one-percentage point increase in the tax rate. In fact, small firms' leverage responds more strongly to tax increases, suggesting that firm size alone is a poor proxy for financing frictions in the European context.[16] Figure 6 illustrates that small firms increase debt levels by a slightly larger magnitude than larger firms following an average tax increase. In monetary terms, this implies an increase in debt of approximately \$170 (\$89,000) for the average small (large) firm, compared with around \$9,000 (\$107,000) for young (old) firms, assuming total assets remain unchanged.

Given that firm age and size are correlated, we jointly account for both by estimating the model with double interactions. The results in Column (3) of Table 3 confirm that financial constraints are primarily associated with firm age rather than firm size. The coefficient of interaction between tax increases and the young-firm indicator variable remains negative and statistically significant, while for small firms the estimated interaction coefficient is positive and also statistically significant. Taken together, these findings are consistent with young firms having more limited collateral and shorter credit histories, together with more profitable opportunities and stronger incentives to borrow. The former means that, compared to old firms, for young firms borrowing constraints kick in at lower leverage ratios, and the latter that young firms will operate closer to or at their borrowing limits. Both restrict the ability to increase leverage following corporate income tax hikes.

By their nature, young firms tend to operate at the technological frontier, experiment with new

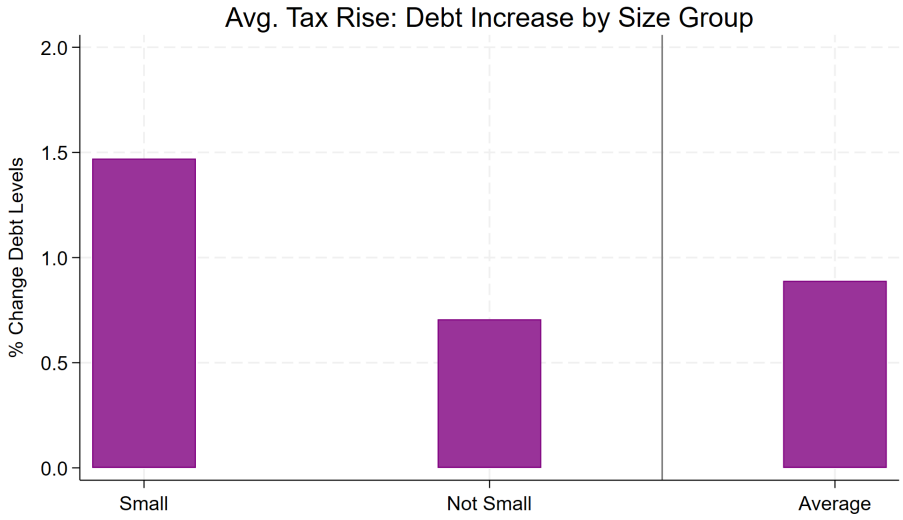
[16] Hurst et al. (2011) find that small business owners are not interested in increasing the scale of their businesses, innovate, or enter new markets and suggest that "the importance of entrepreneurial talent, entrepreneurial luck, and financial frictions in explaining the firm size distribution may be overstated."

Figure 5: Debt Effect of Average Tax Rise By Age



The figure plots the percentage change in debt levels after an average tax increase of 2.1 percentage points for each age group using the coefficients from our benchmark regression with the interaction term of the young dummy (column 1 in table 3). We assume that total assets remain unchanged following the tax increase, in line with Farre-Mensa and Ljungqvist (2016).

Figure 6: Debt Effect of Average Tax Rise By Size



The figure plots the percentage change in debt levels after an average tax increase of 2.1 percentage points for each size group using the coefficients from our benchmark regression with the interaction term of the small dummy (column 2 in table 3). We assume that total assets remain unchanged following the tax increase, in line with Farre-Mensa and Ljungqvist (2016). A firm is small if its total assets are in the lowest quartile of the distribution.

products or business models, and invest more heavily in intangible assets with highly uncertain returns. Although these activities generate potentially large social returns (Jones and Summers 2022), they also expose firms to substantial cash-flow volatility and a higher risk of failure, which makes access to finance more difficult. As a consequence, young firms tend to rely more heavily on venture capital and other forms of long-term risk capital. Therefore, we also examine whether

firms that receive venture capital financing are financially constrained by introducing an interaction of the tax hikes variable with a dummy variable that equals one if the firm has received funding from a venture capital firm (and zero otherwise). Column (4) in Table 3 reports the results, which indicate that venture capital backed firms do not change their leverage in response to tax increases.[17] Therefore, according to the debt test used in this paper, VC-backed firms are more likely to face debt-based financial constraints and to have to resort to internal funds or issue equity, compared to all other firms.

Table 4: Heterogeneity Across Sectors: Innovation Concentration

	(1) Innovation	(2) Innov. × Young	(3) Innov. × Small	(4) Innov. × Young × Small
Rise	0.081*	0.090*	0.065	0.070
	(0.041)	(0.045)	(0.040)	(0.043)
Rise × Young		-0.077***		-0.064***
		(0.018)		(0.017)
Rise × Small			0.061**	0.083*
			(0.026)	(0.042)
Rise × Young × Small				-0.060
				(0.041)
Rise × Innovation	-0.019	-0.040	0.005	-0.000
	(0.043)	(0.046)	(0.049)	(0.047)
Rise × Innovation × Young		0.120**		0.042
		(0.054)		(0.047)
Rise × Innovation × Small			-0.093**	-0.194***
			(0.043)	(0.044)
Rise × Innovation × Young × Small				0.297***
				(0.029)
Obs.	24,338,530	24,338,530	24,338,530	24,338,530
R-squared	0.115	0.115	0.115	0.115
Firm controls	✓	✓	✓	✓
Macro controls	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓

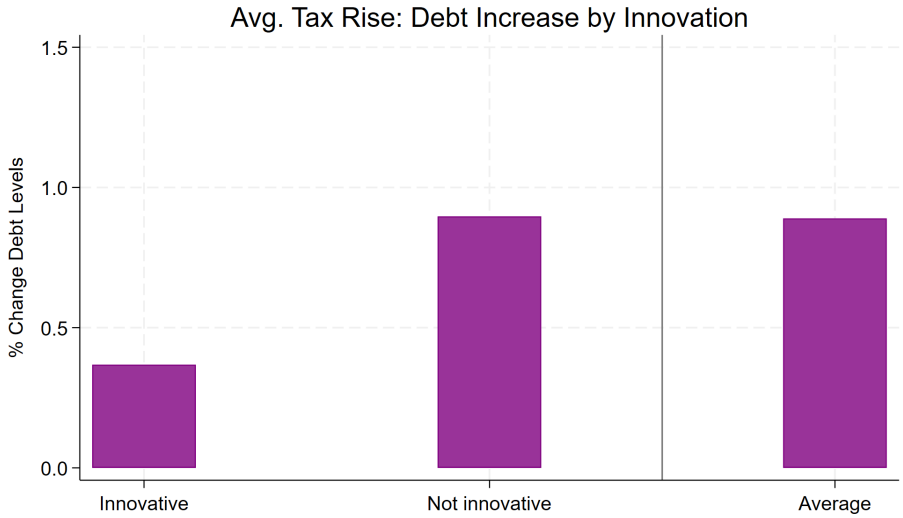
This table reports coefficients from the baseline panel regression for unlisted firms in more and less innovative industry-country pairs. The dependent variable is the change in the firm's long-term leverage ratio in percent. Rise and Cut are continuous indicators of increases or decreases in our composite corporate tax rate measure and the main independent variables. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment rate, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding from CIT. All independent variables are lagged by one period. *Young* and *small* are defined as above. *Innovation* equals 1 if the firm is in a country-industry pair which exhibits a high share (top quartile) of innovative firms according to the Oslo indicator in a given year. The empirical model is estimated in first differences using industry-year fixed effects, firm fixed effects, and robust standard errors clustered at the country level. Significance levels are * p<0.10, ** p<0.05, *** p<0.01.

Venture capital is more directed toward young, innovative, and high-growth firms, and VC-backed firms tend to generate more innovation than comparable non-VC-backed firms (Kortum and Lerner 2000, Chemmanur et al. 2014). Innovative firms therefore provide a natural setting in which to study financial constraints. This is because investment in innovation carries substantial uncertainty about future returns and relies heavily on intangible, firm-specific assets that have limited collateral value, which makes them difficult to finance with debt (Brown et al. 2013). At

[17] Although the benchmark effect of tax increases on leverage remains positive and significant, it is more than offset for venture capital backed firms by a negative interaction term, implying a small and negative combined effect.

the same time, firm innovation (including VC-backed) is a central driver of technological change and aggregate productivity growth (Klette and Kortum 2004, Opp 2019). Given this, we examine whether financing frictions are systematically stronger in innovation-intensive sectors than in other sectors. We do this by introducing in Equation (4) an interaction of *rise* with a dummy variable *Innovation* such that $Innovation_{ijct} = 1$ if firm i is in country c and sector j that ranks in the top quartile of the Oslo indicator in year t (see OECD and Eurostat 2018). We also examine whether firm age and size interact with those frictions.

Figure 7: Debt Effect of Average Tax Rise By Innovation



The figure plots the percentage change in debt levels after an average tax increase of 2.1 percentage points for each innovation group using the coefficients from our benchmark regression with the interaction term for the innovation dummy (column 1 in table 4). We assume that total assets remain unchanged following the tax increase, in line with Farre-Mensa and Ljungqvist (2016).

Column (1) in Table 4 highlights how the leverage response to tax incentives varies with country-sector innovation intensity. The result is that firms operating in more innovation-intensive industries exhibit a weaker leverage response, consistent with tighter financial constraints in innovative activities. Figure 7 illustrates that innovative firms increase debt levels by about a quarter less than firms in less innovative industries: 6 and 8 basis points, respectively.[18] In monetary terms, this implies an increase in long-term debt levels of approximately $97,000 ($76,000) for the average firm in an innovative (less innovative) industry-country pair, assuming total assets remain unchanged.

However, the weaker response of innovation-driven firms is driven by old firms, not young ones.

[18] This is 0.9 percent and 1.4 percent for firms in innovation-driven sectors and the rest, respectively, following an average tax increase.

In Table 4, Column (2), the estimated coefficient of interaction between *Innovation* and *Young* is positive and statistically significant, with the leverage response of innovative young firms being higher (21 basis points) than that of non-innovative old firms (9 basis points), non-innovative young firms (1.3 basis points), and innovative old firms (5 basis points). This indicates that young firms in innovative sectors have the best access to finance while old firms in innovate sectors have the worst.

Table 4, Column (3) shows that small firms in innovative sectors exhibit a significantly weaker leverage response, suggesting that innovation amplifies financing constraints for small firms. Finally, the estimates of a specification with a four-way interaction of *Innovation*, *Young*, and *Small* yields a paradoxical result for firms in innovation-driven sectors: as per Column (4), small, old firms are the most financially constrained since their leverage ratio decreases by 4 basis points in response to a 1 percentage point hike in the corporate income tax rate, while young, small firms are the least financially constrained as their leverage ratio increases by 17 basis points. In the other sectors of the euro area economy, young firms are generally more constrained than old ones, regardless of size. A possible explanation for this last set of results is that, in the past, firms in innovation-intensive country–industry pairs may have faced a bottleneck driven by lack of access to debt finance. This means that, unlike young and innovative firms in the sample, they were not able to finance the investments needed during their scale-up phase, in a "lost generation" effect.

4.4 Robustness

In this section, we assess the robustness of our findings. We undertake three robustness exercises. First, we use as the dependent variable the change in long-term leverage from *t–2* to *t*, instead of year-on-year to rule out anticipatory effects of the tax change. Second, we expand the set of firm-level controls used in the benchmark specification (1). The additional controls are depreciation and amortization charges, and the ratio of tangible fixed assets to total assets (a measure of a firm's asset tangibility), which are relevant in the context of corporate taxation and innovation intensity. Third, we conduct a placebo test to ensure that the changes in leverage ratios are driven by the tax changes and are not spurious. For the placebo test, we shift the tax changes variables forward by two periods, to *t+1*. Table B.14 reports the results.

Column (1) of Table B.14 shows that the alternative horizon does not change the direction

results: the average private firm is still unconstrained because it increases leverage in response to a tax hike, though the response is no longer statistically significant. The estimate is somewhat smaller than what we have in Column (1) of Table 2, with a one percentage point increase in the corporate income tax rate now being associated with a 5.9 basis point increase in leverage over the two-year outcome window, equivalent to about 3 basis points over one year.

Our results are also robust to the inclusion of additional firm-level controls, as the results in Column 2 of Table B.14 are similar to those in Column (1) of Table 2. This also holds for the regressions including interaction terms with young and small (available from the authors). Results are also robust to the use of controls in first differences instead of levels (not reported), as well as to using country fixed effects instead of firm fixed effects (Table B.12). We have also checked whether our results were robust to clustering our standard errors at the country-year level and they are mostly unchanged, see Table B.13.

Third, the results from the placebo test, shown in Column (3) of Table B.14, show that our results are not driven by unobservable firm characteristics, confounding variables, or spurious treatment assignment, but instead come from the proposed mechanism. Since the coefficient we estimate is no longer statistically significant, we conclude that the tax changes we identify drive the changes in long-term leverage. All results hold for the full sample that includes the listed firms too.

5 Conclusion

This paper provides new evidence regarding financial constraints among euro area firms. It does this by exploiting exogenous changes in corporate tax rates as quasi-experimental shocks to firms' demand for debt. Leveraging a large, harmonized dataset covering over 7 million predominantly private firms across 20 euro area countries, we implement a "debt test" *à la* Heider and Ljungqvist (2015) to assess whether firms are financially constrained, sidestepping the limitations of traditional proxies based on financial ratios.

We have five main findings. First, the average euro area firm is not financially constrained, as it raises long-term leverage significantly following tax increases. Quantitatively, unlisted firms—which dominate the European corporate sector—increase long-term leverage by approximately 8 basis points per percentage-point rise in the corporate tax rate, corresponding to an average debt increase

of around $82,300, or 0.2 percent of assets.

Second, public and private firms are similarly unconstrained in their access to debt finance. Most other studies consider either public or private firms, and do not test for differences in access to finance by either type of firm. Since private firms are more opaque than listed firms (because of disclosure requirements for listing), they are more likely to suffer from information asymmetry problems when accessing finance. However, we find that this does not seem to hinder their ability to tap debt. This could be because asymmetric information among euro area firms is not as severe as expected, or because their lenders (banks) specialize in solving this type of problem.

Third, this average response conceals important heterogeneity across firm types. Young firms exhibit significantly weaker leverage responses than older firms, consistent with tighter borrowing limits arising from shorter credit histories or limited collateral. Given that young firms play a disproportionate role in driving productivity growth and innovation, this result carries important policy implications. By their nature, young firms tend to operate at the technological frontier, experiment with new products or business models, and invest heavily in intangible assets whose returns are uncertain and often back-loaded. Although these activities generate potentially large social returns, they also expose firms to substantial cash-flow volatility and a higher risk of failure. At the same time, young firms typically lack established credit histories, accumulated retained earnings, and tangible collateral, which makes access to external finance (i.e., debt) more difficult. As a consequence, young firms are less able to adjust borrowing in response to changes in debt incentives and rely more heavily on internal funds or long-term risk capital, such as venture capital. However, firms that have received VC funding also appear to be more constrained.

Fourth, firm size alone is a poor proxy for financing frictions. We find that small firms increase their leverage by significantly more than larger firms after a tax rise. Standard constraint indexes such as the Whited-Wu index fail to identify constrained firms in this predominantly private, unlisted setting in Europe, underscoring the value of the debt test.

Fifth, financing frictions in innovation-intensive environments follow a non-linear pattern that points to a scale-up bottleneck in Europe. Small mature firms in innovative sectors are the most constrained, while young innovative firms, including young and small firms, retain the ability to expand borrowing when incentives strengthen. Taken together, our results support the view that debt-financing constraints are not the dominant barrier for the average euro area firm, but also

that they may remain important for specific groups like young, high-growth firms, and mature firms that are small and in innovation-driven sectors. These are also the firms for which external finance is more important to grow and to increase productivity.

Although our paper covers a large cross-section of firms and uses a well-identified approach to measure financial constraints, there are some important caveats to note. First, this paper considers only debt-financing constraints, but financial constraints are multidimensional. In addition to debt, firms may not be able to access equity, or they may not be able to generate sufficient internal funds to finance growth. These constraints in accessing other sources of finance may bind for different types of firms or bind at different times in the firm's life cycle. For example, our results are consistent with those of Hoberg and Maksimovic (2015), who find that larger and younger firms are debt-constrained, small firms are equity-constrained, and medium-size and medium-age firms have trouble accessing private placement finance. Their findings, with ours, support the view that financial constraints may be more pervasive than what is suggested by the debt test that we use in this paper.

Second, we find that private and public firms seem to be similarly constrained in accessing debt finance, which we understand as suggestive of information asymmetries playing a less important role than financial distress (and bankruptcy costs) in explaining financial constraints for euro area firms. However, information asymmetries are important for firms wanting to access equity markets (Hoberg and Maksimovic 2015), but we do not test that here.

Third, our findings about access to debt finance by firms in innovation-driven sectors can provide useful stylized facts for the discussions about Europe's scale-up gap, but the debt test we use does not directly measure access to venture capital, equity, or other forms of risk capital. Still, our findings highlight the importance of considering how different forms of external finance support firm growth at different stages of the firm life cycle.

Fourth, although the debt test helps with endogeneity and supports a causal interpretation of our findings, it does not fully solve the problem. For example, the interpretation that financial distress is the main factor constraining access to finance can be reversed because high leverage can signal easy access to debt finance that, over time, causes firms to experience financial distress (Buehlmaier and Whited 2018).

Finally, we conduct the debt test using firms' financial returns, which may obscure the tax incentives for firms to borrow compared to data from tax returns (see Devereux et al. 2018), but since we preferentially use unconsolidated balance sheet data and most of the firms in our sample are small and private, the measurement error introduced by these data may not be as problematic as in other studies (e.g., Heider and Ljungqvist 2015). Furthermore, the attenuation bias induced by the use of balance sheet data works in our favor as it likely means that we still overestimate the severity of financial constraints among euro area firms.

Appendix A: Data

We utilize firm-level data from Bureau van Dijk's Orbis Historical, merging data on firmographics (including firm characteristics like name, age and sector) with balance sheet financials. The following sets out the precise cleaning procedure, which broadly follows Kalemli-Özcan et al. (2024).

Orbis Firmographics Dataset

We take the following steps to eliminate duplicates, with the goal of obtaining one observation per firm (as identified by its unique BvD ID number). We first drop duplicates in terms of **all variables** in the sample. Second, we aim to eliminate **duplicates in terms of National ID, National VAT, EU VAT and ISIN number**, to then break this down further with every step. As such, of the duplicate rows, we keep the ones with non-missing National VAT numbers. For these rows where National ID, National VAT, EU VAT and ISIN number are identical, we drop the ones with missing address types. If rows are still duplicate in all four identifiers but exhibit two different address types, we keep the one equal to "Registered address" as it is the most common address type in our sample. This eliminates all duplicates in terms of the four identifiers above. Third, we eliminate duplicates in terms of **National ID, National VAT, EU VAT** by dropping the rows with a missing ISIN number. Next, we forward and backward fill the European VAT number where missing. We aim to preserve as much variation in national identifiers as possible, such that we drop rows where some non-missing identifiers are identical:

- Drop if duplicate in BvD ID and National VAT and National VAT equals National ID
- Drop if duplicate in BvD ID and National VAT and EU VAT equals National ID
- Drop if duplicate in BvD ID and National VAT and EU VAT is missing
- Drop if duplicate in BvD ID and EU VAT and National VAT is missing
- Drop if duplicate in BvD ID and EU VAT and EU VAT equals National VAT

Finally, for the remaining duplicates in terms of **BvD ID number**, we drop rows where both National and EU VAT numbers are missing. As a last step, we drop duplicate BvD ID observations

which have a missing EU VAT number. Our resulting sample has unique observations by firm (BvD ID number) with maximum variation in identifiers.

Orbis Financials Dataset

For the balance sheet variables, we aim for unique firm-financial year observations. Therefore, we first establish unique observations in terms of **BvD ID, financial year, consolidation code and filing type** by keeping the latest observation, i.e. a reporting month closer to December, and a reporting day closer to 31. Second, to get unique observations in terms of **BvD ID, year, and consolidation code**, we drop duplicate rows of these three variables where the filing type is local registry filings in favor of annual reports. This is because local registry filings are more often coded as limited financials. Third, to get unique **BvD ID and year** observations, we deal with consolidation codes. Following Kalemli-Özcan et al. (2024), we keep unconsolidated over consolidated accounts to avoid double counting. Within consolidated accounts, we keep those where the company headquarters does not present an unconsolidated account. Within unconsolidated accounts, we keep those where the company headquarter does not present a consolidated account. Next, we treat same-year reporting by keeping the observation with the reporting month closer to December. Similarly, for same-month reporting, we keep the reporting day closer to the end of the month. Finally, for BvD ID-year observations which differ only in terms of original reporting units, we keep the observations in "units" as this is the most frequent reporting unit.

Having eliminated duplicates, we eliminate likely reporting errors, employing basic data cleaning following Kalemli-Özcan et al. (2024). As such, we drop firms with negative total assets, sales, fixed assets, or tangible fixed assets. Furthermore, we drop observations with negative employees or more than 2 million employees. We also drop observations where total assets *and* operating revenue *and* sales *and* employment are jointly missing. Observations with a negative leverage ratio, computed as (long-term debt + loans)/ total assets, are dropped. Finally, we construct an indicator for inconsistent balance sheets by comparing total assets to the sum of shareholder funds, current and non-current liabilities. This ratio should equal 1 for consistency, and we drop observations with a ratio below the 1st or above the 99th percentiles.

Orbis Merged Firmographics - Financials Dataset

We are using an inner join between the two above datasets, keeping only firms that have both Firmographics and Financials data. Further cleaning steps on the merged data include dropping a few observations with missing NACE2 main sector classifications, and observations which belong to sectors K (Financial and insurance activities), O (Public administration and defense), U (Extraterritorial organizations) and T (Activities of households as employers). In addition, we drop firms where the reporting year predates the year of incorporation (negative age). We also drop observations with missing or zero values of total assets, and with negative long-term debt, loans, or R&D expenses while keeping observations with missing values. We exclude firms which change original reporting units over time when the transformation between units is implausible. For those which report in thousands or millions and do not switch units over time, we multiply the financial values by 1000 or 1000000, respectively, to harmonize the dataset to be in units.

As these variables are crucial for our analysis, we only keep observations with non-missing values for age, total assets, listed status and fixed assets. We then trim long-term leverage, which is our variable of interest, as well as age and total assets, at the 1st and 99th percentiles.

We notice that the country distributions of some variables central to our analysis are not representative of our full sample because reporting requirements vary substantially across countries in Orbis. In particular, long-term debt is not consistently reported for all firms and countries. To maximize country coverage, we therefore impute missing values of long-term leverage, sales, and cash flow using linear regressions estimated on observations with non-missing values.

For long-term leverage, the predictors include country, sector, age, size category, listed status, total assets, asset growth, tangible fixed assets, investment, and cash and cash equivalents. This procedure increases the number of observations for which long-term leverage is available from 39.5 million to 47.9 million. Table B.10 compares the distribution of the original and imputed long-term leverage series and shows that the two distributions are very similar.

As a robustness check, we re-estimate our benchmark specification using the non-imputed long-term leverage measure. The results remain qualitatively unchanged and, if anything, become slightly larger in magnitude than in the baseline specification (see Table B.11).

Since our data are reported in nominal U.S. dollars, we convert the financial variables to real

values. We do this by converting all financial variables to the original exchange rate (Euro), deflating by the original currency deflator at the industry-level, and then converting back to U.S. dollars.

Appendix B: Additional Tables

Table B.1: Range of Estimates for the Effect of a 1 p.p. Tax Increase on Firm Leverage

Region	**Firm Type**	**Estimated Effect (bps per 1 p.p. ↑)**	**Reference**
USA	Public & Private	**15 bps** (aggregate)	Fleckenstein et al. (2020)
USA	Public	**38.7 to 40.5 bps**	Heider and Ljungqvist (2015)
USA	Public	**No effect**	Graham et al. (2015)
USA	Small Private	**Negative Sensitivity** (Borrowing falls)	Ivanov et al. (2025)
United Kingdom	Private	**76 to 140 bps**	Devereux et al. (2018)
European countries	Private with ownership data	**25 bps**	Krämer (2015)
OECD countries (29)	Public	≈ **40 bps** (converted from 0.41 elasticity)	Faccio and Xu (2015)
Most Countries (93)	Private	**16.3 bps** (fully deductible interest)	Faccio and Xu (2025)
International	Multinational Affiliates	≈ **45 bps** (converted from 0.28 elasticity)	Desai et al. (2004)

Note: Conversions for Faccio and Xu (2015) use mean leverage (36.73 percent) and mean tax (37.75 percent). Conversions for Desai et al. (2004) use mean affiliate leverage (55.18 percent) and mean country tax (34.31 percent).

Table B.2: Summary Statistics: All Euro Area Firms

All firms (listed and unlisted).	p25	Mean	Median	p75	SD	Obs
Age	6.00	13.67	11.00	19.00	10.36	47.94
Employees	2.00	14.33	4.00	10.00	237.87	30.41
Total assets (mio.)	0.14	49.16	0.50	1.65	498.98	47.94
Asset growth (%)	-15.10	-0.64	-1.29	14.60	39.08	38.20
EBITDA (mio.)	0.00	3.80	0.03	0.11	4590.86	38.58
Δ Cash/Assets (pp)	-3.56	0.31	0.02	4.29	15.10	36.93
Tangible FA (TFA, mio.)	0.01	10.94	0.05	0.32	202.87	46.14
TFA/Assets (%)	2.58	24.74	12.75	39.10	631.98	46.14
Investment rate (%)	-5.46	-4.53	-0.58	2.13	1060.36	47.94
Sales (imp., mio.)	0.13	179.26	0.58	4.89	1772.66	47.94
Sales growth (imp., %)	-16.68	16.83	-0.51	17.36	121.80	47.94
Turnover (mio.)	0.11	77.70	0.38	1.43	1806.35	40.74
ROA (%)	-0.90	3.79	2.53	9.94	19.43	40.57
Long-term debt (levels, mio.)	0.00	10.98	0.01	0.18	213.17	39.54
Long-term leverage (imp., %)	0.00	18.88	9.80	29.13	25.08	47.94
Δ Long-term leverage (imp., %)	-1.77	-0.22	0.00	0.00	14.54	38.20
Leverage (%)	0.00	21.48	8.35	32.98	1464.78	36.81
WW index	-2.16	-1.52	-0.77	0.43	5.65	47.94

This table shows the summary statistics for key characteristics (age, number of employees), balance sheet variables with a particular focus on the debt-related variables, and one traditional index of financial constraints from the literature for the full sample of listed and unlisted firms. FA refers to fixed assets; TFA to tangible fixed assets; ROA to the return on assets. WW index refers to the Whited-Wu (2006) index, which takes higher values for more constrained firms. Observations are reported in millions.

Table B.3: Summary Statistics: Tax and Macro Variables

	Min	Mean	p50	Max	SD	Obs (mio.)
Corporate Income Tax Rate (Level)	12.50	26.97	27.50	44.43	6.10	47.94
Tax Rise	0.00	0.16	0.00	10.00	0.85	47.94
Tax Cut	-15.83	-0.61	0.00	0.00	1.58	47.94
GDP Gr.	-16.04	1.17	1.57	24.62	3.68	47.94
CPI	-1.68	2.17	1.74	19.45	2.40	47.94
Unemployment Rate	2.22	10.54	9.23	27.48	4.81	47.94
Gov. Expenditure (log)	0.64	5.87	6.22	7.57	1.31	47.92
Gov. Revenue ex CIT (log)	0.38	5.73	6.09	7.46	1.31	47.92
10y Yield Spread	-1.22	1.07	0.66	21.89	1.36	46.17
2y Yield	-0.80	1.57	0.72	174.44	6.30	46.50
Δ GDP Gr.	-23.39	0.23	-0.12	20.95	5.31	38.20
Δ CPI	-11.99	0.29	-0.08	14.96	2.43	38.20
Δ Unempl. Rate	-4.38	-0.11	-0.35	9.80	1.46	38.20
Δ Gov. Expenditure (log)	-0.29	0.03	0.03	0.31	0.04	38.19
Δ Gov. Revenue ex CIT (log)	-0.17	0.03	0.03	0.27	0.04	38.19
Δ 10y Yield Spread	-13.60	0.02	0.00	9.02	0.86	36.76
Δ 2y Yield	-158.65	-0.20	-0.15	129.54	4.27	37.03

This table shows the summary statistics for our composite corporate tax measure, the level and the changes of our macroeconomic control variables for the full sample of listed and unlisted firms. Observations are reported in millions.

Table B.4: Summary Statistics: Young vs. Old Euro Area Firms

	Mean		Median		Std. Dev.		Obs. (Mio.)	
Variable	Old	Young	Old	Young	Old	Young	Old	Young
Age	17.57	4.18	15.00	4.00	9.92	1.30	33.94	13.98
Employees	16.45	8.25	5.00	3.00	255.19	181.09	22.32	8.08
Total assets (mio.)	57.94	27.35	0.66	0.24	551.00	337.00	33.94	13.98
Asset growth (%)	-2.00	5.00	-2.00	2.00	36.00	49.00	29.62	8.56
EBITDA (mio.)	4.78	1.59	0.04	0.02	5500.00	355.00	26.79	11.77
TFA (mio.)	12.92	5.98	0.073	0.022	223.00	141.00	32.80	13.33
TFA/Assets (%)	26.00	23.00	14.00	11.00	749.00	27.00	32.80	13.33
Intangible FA (mio.)	2.20	2.18	0.00	0.00	45.50	44.20	32.80	13.33
Intangible FA/Assets (%)	3.17	6.39	0.00	0.00	11.02	16.22	32.80	13.33
Investment Rate (%)	-4.91	-3.60	-0.73	-0.21	1,230.22	425.81	33.94	13.98
Sales (imp., mio.)	211.00	102.00	0.78	0.33	2030.00	894.00	33.94	13.98
Sales growth (imp, %)	6.71	41.37	-1.43	4.10	103.18	155.42	33.94	13.98
Turnover (mio.)	92.23	44.15	0.45	0.26	2090.00	843.00	28.31	12.42
ROA (%)	3.30	4.92	2.25	3.49	17.25	23.66	28.23	12.33
Long-term debt (mio.)	12.40	7.50	0.02	0.00	225.00	179.00	27.94	11.59
Long-term leverage (imp., %)	19.00	20.00	10.00	9.00	24.00	26.00	33.94	13.98
Leverage (%)	21.00	22.00	9.00	7.00	1751.00	72.00	25.74	11.06

This table shows the summary statistics for firms we classify as young or old. It includes key characteristics (age, number of employees), balance sheet variables with a particular focus on the debt-related variables, and one traditional index of financial constraints from the literature for unlisted firms. FA refers to fixed assets; TFA to tangible fixed assets; ROA to the return on assets.

Table B.5: Summary Statistics: Small vs. Not Small Euro Area Firms

	Mean		Median		Std. Dev.		Obs. (Mio.)	
Variable	Not Small	Small	Not Small	Small	Not Small	Small	Not Small	Small
Age	14.88	10.04	12.00	7.00	10.69	8.28	35.94	11.98
Employees	17.66	2.76	5.00	2.00	269.49	40.83	23.49	6.91
Total assets (mio.)	65.33	0.06	0.86	0.06	574.00	0.04	35.94	11.98
Asset growth (%)	2.00	-9.00	-0.00	-6.00	34.00	52.00	29.40	8.78
EBITDA (mio.)	5.10	0.01	0.05	0.00	5,320.00	0.20	28.74	9.82
TFA (mio.)	14.29	0.01	0.11	0.01	232.00	0.02	35.22	10.91
TFA/Assets (%)	26.00	21.00	13.00	11.00	723.00	25.00	35.22	10.91
Intangible FA (mio.)	2.87	0.00	0.00	0.00	51.60	0.00	35.22	10.91
Intangible FA/Assets (%)	4.26	3.62	0.00	0.00	12.98	12.30	35.22	10.91
Investment Rate (%)	-1.82	-12.64	-0.46	-1.05	499.74	1,936.10	35.94	11.98
Sales (imp., mio.)	229.00	29.82	1.17	0.09	2,040.00	162.00	35.94	11.98
Sales growth (imp, %)	15.59	20.52	-0.30	-2.11	121.72	121.98	35.94	11.98
Turnover (mio.)	107.00	0.13	0.73	0.09	2,120.00	0.25	29.62	11.11
ROA (%)	3.99	3.24	2.36	3.44	15.10	28.29	29.96	10.60
Long-term debt (mio.)	13.97	0.01	0.04	0.00	240.00	0.02	31.01	8.52
Long-term leverage (imp., %)	19.00	18.00	10.00	9.00	25.00	25.00	35.94	11.98
Leverage (%)	22.00	19.00	11.00	0.00	1,659.00	190.00	28.57	8.22

This table shows the summary statistics for firms we classify as small or not small. It includes key characteristics (age, number of employees), balance sheet variables with a particular focus on the debt-related variables, and one traditional index of financial constraints from the literature for unlisted firms. FA refers to fixed assets; TFA to tangible fixed assets; ROA to the return on assets.

Table B.6: Number of observations by country

Country	Total obs.	Obs. in tax-rise years	Obs. in tax-cut years
Italy	7,863,794	614,986	2,776,613
Spain	7,210,153	0	1,450,977
France	3,339,987	647,254	882,388
Portugal	2,872,028	758,434	227,906
Finland	1,132,202	0	228,734
Slovakia	616,617	38,740	95,070
Belgium	543,717	36,638	86,675
Slovenia	510,188	55,134	52,960
Croatia	315,027	0	49,716
Greece	195,149	26,848	50,757
Latvia	183,629	0	0
Germany	133,786	6,684	18,822
Ireland	61,021	0	825
Netherlands	37,815	0	4,402
Austria	9,417	0	1,024
Malta	6,444	0	954
Cyprus	9	0	0
Total	25,030,983	2,184,718	5,927,823

Table B.7: Number of observations by year

Year	Total obs.	Obs. in tax-rise years	Obs. in tax-cut years
2002	370,116	0	169,485
2003	405,671	0	111,026
2004	474,749	1,142	94,156
2005	583,318	0	171,308
2006	586,057	0	134,405
2007	741,998	0	129,435
2008	868,405	0	274,487
2009	1,145,909	0	678,627
2010	1,271,892	0	0
2011	1,365,583	0	11,694
2012	1,421,971	442,803	14,949
2013	1,450,255	162,437	56,740
2014	1,471,585	288,707	52,960
2015	1,503,209	0	103,338
2016	1,556,869	13,214	611,679
2017	1,612,306	0	571,361
2018	1,668,426	186,120	663,494
2019	1,717,014	217,122	745,881
2020	1,761,311	0	620,138
2021	1,784,278	836,535	240,722
2022	1,270,061	36,638	471,938
Total	25,030,983	2,184,718	5,927,823

Table B.8: Corporate income tax rises

Country	Year of tax change	Year of regression	Obs.	Size of rise (pp)
Germany	2003	2004	1,142	1.58
France	2011	2012	279,935	1.67
Portugal	2011	2012	162,868	2.00
Portugal	2012	2013	162,437	3.00
France	2013	2014	236,333	1.90
Greece	2013	2014	13,634	6.00
Slovakia	2013	2014	38,740	4.00
Greece	2015	2016	13,214	3.00
France	2017	2018	130,986	10.00
Slovenia	2017	2018	55,134	2.00
Germany	2018	2019	5,542	0.01
Portugal	2018	2019	211,580	1.80
Italy	2020	2021	614,986	0.86
Portugal	2020	2021	221,549	0.07
Belgium	2021	2022	36,638	0.10

Table B.9: Corporate income tax cuts

Country	Year	Obs.	Cut (pp)	Country	Year	Obs.	Cut (pp)
France	2001	103,381	−1.33	Slovenia	2013	52,960	−1.00
Italy	2001	66,104	−1.00	Finland	2014	62,582	−4.50
France	2002	100,396	−1.00	Slovakia	2014	40,756	−1.00
Netherlands	2002	118	−0.50	Spain	2015	418,963	−2.00
Portugal	2002	10,512	−2.00	Portugal	2015	192,716	−2.00
Belgium	2003	506	−6.00	Spain	2016	425,903	−3.00
Ireland	2003	825	−3.50	France	2016	145,458	−3.57
Italy	2003	92,825	−2.00	Croatia	2017	49,716	−2.00
Germany	2004	2,584	−1.58	Italy	2017	559,464	−3.50
Italy	2004	144,046	−1.00	Slovakia	2017	54,314	−1.00
Portugal	2004	24,678	−5.00	Belgium	2018	43,684	−3.95
Austria	2005	9	−9.00	France	2018	122,026	−9.27
Finland	2005	27,554	−3.00	Italy	2018	579,217	−0.29
France	2005	106,612	−0.50	Malta	2018	954	−4.54
Netherlands	2005	230	−3.00	Greece	2019	20,040	−4.62
France	2006	129,137	−0.50	Italy	2019	600,098	−0.02
Netherlands	2006	298	−1.90	Austria	2020	1,015	−0.50
Spain	2007	274,039	−2.50	Belgium	2020	42,485	−3.76
Netherlands	2007	448	−4.10	Germany	2020	6,177	−0.29
Germany	2008	8,987	−10.55	Finland	2020	81,858	−0.32
Spain	2008	332,072	−2.50	France	2020	109,187	−2.23
Italy	2008	337,568	−5.50	Germany	2021	1,074	−1.62
Greece	2010	11,694	−1.00	France	2021	66,191	−3.28
Greece	2011	11,641	−4.00	Greece	2021	7,382	−1.85
Netherlands	2011	3,308	−0.50	Italy	2021	397,291	−5.80
Finland	2012	56,740	−1.50				
				Total		5,927,823	

Notes: Year refers to the year of the tax change; the corresponding regression year is the following year. Size of cut in percentage points.

Table B.10: Summary statistics: long-term leverage

	Min	P25	Mean	Median	P75	Max	SD	Obs
Long-term leverage	0.000	0.000	0.172	0.028	0.252	2.121	0.271	39.54
Long-term leverage, imputed	0.000	0.000	0.189	0.098	0.291	2.121	0.251	47.94
Δ Long-term leverage	−2.121	−0.026	−0.004	0.000	0.000	2.121	0.149	30.76
Δ Long-term leverage, imputed	−2.121	−0.018	−0.002	0.000	0.000	2.121	0.145	38.20

Notes: Obs reported in millions.

Table B.11: Leverage Response to Tax Changes, Non-imputed Leverage Ratio

Listed Status	(1) Unlisted	(2) Listed	(3) All
Rise	0.093**	0.152	0.093**
	(0.040)	(0.146)	(0.040)
Rise × Listed			0.061
			(0.123)
Cut	-0.039	0.255**	-0.039
	(0.034)	(0.103)	(0.034)
Cut × Listed			0.122
			(0.100)
Observations	20,234,171	4,205	20,242,605
R-squared	0.138	0.220	0.138
Firm controls	✓	✓	✓
Macro controls	✓	✓	✓
Industry-Year FE	✓	✓	✓
Firm FE	✓	✓	✓

This table reports coefficients from the benchmark panel regression using non-imputed changes in long-term leverage for the full sample and for sub-samples based on the firm's listing status. Rise and Cut are continuous indicators of increases or decreases in our composite corporate tax rate measure and the main independent variables. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment rate, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding from CIT. All independent variables are lagged by one period. The empirical model is estimated in first differences using industry-year fixed effects in all columns, firm fixed effects in columns (1) and (2), and robust standard errors clustered at the country level. Significance levels are * p<0.10, ** p<0.05, *** p<0.01.

Table B.12: Different Assumptions on Heterogeneity: Leverage Response to Tax Changes

	(1) Country FE	(2) Country FE	(3) Firm FE	(4) Firm FE
Rise	0.108**	0.098**	0.088**	0.080*
	(0.041)	(0.043)	(0.040)	(0.041)
Cut	-0.043**	-0.036	-0.009	-0.007
	(0.018)	(0.028)	(0.019)	(0.032)
Observations	26,308,249	25,030,983	25,598,461	24,347,346
R-squared	0.005	0.006	0.113	0.115
Firm controls	✓	✓	✓	✓
Macro controls		✓		✓
Industry-Year FE	✓	✓	✓	✓
Firm FE			✓	✓
Country FE	✓	✓		

This table reports coefficients from the benchmark panel regression for different combinations of fixed effects and macroeconomic controls. The dependent variable is the change in the firm's long-term leverage ratio. Rise and Cut are continuous indicators of increases or decreases in our composite corporate tax rate measure and the main independent variables. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment rate, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding from CIT. All independent variables are lagged by one period. The empirical model is estimated in first differences using industry-year fixed effects and robust standard errors clustered at the country level. Significance levels are * p<0.10, ** p<0.05, *** p<0.01.

Table B.13: Leverage Response to Tax Changes, Clustered Standard Errors by Country and Year

Sample	(1) Unlisted	(2) Listed	(3) All	(4) Unlisted	(5) All
Rise	0.080*	0.129	0.080*	0.075	
	(0.045)	(0.134)	(0.045)	(0.043)	
Rise × Listed			0.091		
			(0.168)		
WW				-0.077	
				(0.056)	
Rise × WW				0.036	
				(0.038)	
Rise (small)					0.627*
					(0.347)
Rise (medium)					0.423**
					(0.194)
Rise (large)					0.293*
					(0.152)
Cut	-0.007	0.273***	-0.007	0.001	
	(0.034)	(0.086)	(0.034)	(0.034)	
Cut × Listed			0.119		
			(0.158)		
Cut × WW				-0.026	
				(0.017)	
Cut (small)					-0.146
					(0.230)
Cut (medium)					0.497
					(0.398)
Cut (large)					-0.087
					(0.111)
Observations	24,338,530	4,280	24,347,346	24,338,530	24,347,346
R-squared	0.115	0.221	0.115	0.115	0.115
Firm controls	✓	✓	✓	✓	✓
Macro controls	✓	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓

This table reports coefficients from benchmark panel regressions of the change of firms' long-term leverage ratios on corporate tax changes. Columns (1)–(4) report results for the unlisted, listed, full, and unlisted-firm samples, respectively. Note that the difference in sample sizes between the sum of columns (1) and (2), and column (3) is due to the inclusion of delisted firms in (3), and a different number of singleton observations. Column (3) includes interactions with listing status, and column (4) includes interactions with the Whited-Wu indicator. Column (5) replaces the continuous rise and cut measures with indicators for small, medium, and large tax changes based on terciles of the respective distributions. Specifically, for rises and cuts separately, we split the distribution of tax changes into terciles of the country-year distributions of rises and cuts, and construct three indicators per direction, giving six mutually exclusive dummies that enter the regression jointly. Country-years with no tax change are the omitted baseline, so each coefficient measures the average leverage response to a tax change of the relevant size class relative to no change. Firm-level controls include return on assets and total asset growth. Macroeconomic controls include GDP growth, CPI, unemployment, 2-year yields, 10-year yield spreads, log government expenditure, and log government revenue excluding corporate income taxes. All independent variables are lagged by one period. The models include industry-year and firm fixed effects, with robust standard errors clustered at the country-year level. Significance levels are * p<0.10, ** p<0.05, and *** p<0.01.

Table B.14: Robustness Checks

	(1) Lev. Ratio $\Delta(t-2,t)$	(2) Firm Controls	(3) Placebo
Rise	0.059	0.091**	
	(0.042)	(0.041)	
Cut	0.000	-0.015	
	(0.039)	(0.036)	
Rise (F1)			-0.045
			(0.075)
Cut (F1)			0.051
			(0.077)
Observations	24,338,530	21,213,048	19,894,103
R-squared	0.206	0.119	0.110
Firm controls	✓	✓	✓
Macro controls	✓	✓	✓
Industry-Year FE	✓	✓	✓
Firm FE	✓	✓	✓

This table reports robustness checks for the benchmark specification. Column (1) defines the change in long-term leverage over the period from t-2 to t. Column (2) augments the benchmark specification with additional firm-level controls. Column (3) reports a placebo test in which the tax-change variables are shifted forward to t+1. Rise and Cut denote continuous increases and decreases, respectively, in the composite corporate tax rate measure. All specifications include the stated firm-level and macroeconomic controls, industry-year fixed effects, and firm fixed effects. Robust standard errors are clustered at the country level. Significance levels are * p<0.10, ** p<0.05, *** p<0.01.

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PUBLICATIONS

Are European Firms Financially Constrained?
Working Paper No. WP/2026/235